(a)

(b)

Redshift z

Deceleration parameter $q(z)$

$E(z) = H(z)/H_0$

Models

- Hippopede: $\vartheta = 0^\circ$ (hyperconical)
- Hippopede: $\vartheta = 15^\circ$
- Hippopede: $\vartheta = 30^\circ$
- Hippopede: $\vartheta = 45^\circ$
- Hippopede: $\vartheta = 60^\circ$
- Flat ΛCDM + radiation ($\Omega_m = 0.333$)
- de Sitter: $q = -1$

Data

- Cosmic chronometers, H/H_0
- Observed q_0 (CC+Pantheon+SH0ES+BAO; Myrzakulov et al. 2025)
- Observed q_0 (FRB+SNe cosmography; Gao et al. 2024)
- Observed z_t (CC+Pantheon+SH0ES+BAO; Myrzakulov et al. 2025)
- Observed z_t ($H(z)$ compilation; Hu et al. 2025)
- Tangent $E = 1 + (1 + q_0)z$, Myrzakulov et al. 2025
- Tangent $E = 1 + (1 + q_0)z$, Gao et al. 2024

GaPP CC+Pantheon+

- GaPP median
- GaPP 1σ
- GaPP 2σ
- GaPP 3σ
- GaPP extrapolation ($z < 0$)